Rui Qi

Robotics and interactive simulation

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Robotics portfolio | Game portfolio

Education

University of Utah | B.S. in Games

Aug 2022 - Apr 2026 | Degree conferred April 30, 2026 | GPA 3.880/4.000

Selected coursework: Programming for All 1 and 2; Introductory Statistical Inference; Game Engine Fundamentals; Traditional Game Development. Dean's List, multiple semesters.

Robotics Project

Visual Grasp Verification and Failure Recovery | Independent project | Oct 2026

- Defined task goals, tested demonstrations and reported missed grasps, empty transport and failed retries. Questioned an unnecessary obstacle push, prompting alternative-grasp comparisons and a stacked-object task.
- The simulation combines color/depth localisation, programmed grasping and extraction, visual payload checks, and recovery after acquiring the wrong object. Current recovery logic is programmed, not a newly trained policy.
- Current workflow and a relocate-first baseline each completed 7/7 development layouts. Of six stacked cases, two used direct extraction and four required recovery; mean task times were 14.64 vs 14.32 simulated seconds. No untouched final test yet.
- A separate small neural policy trained with PPO for 96 episodes selected among three grasp strategies. On 12 new development-validation layouts, policy and rules each completed 10/12; the policy was slower (12.51 vs 11.03 simulated seconds).

Role: task direction, hands-on failure feedback, comparison requirements and result review. Implementation, debugging and experiment automation used extensive AI assistance.

Relevant Experience

Qainbow LLC | Technical / Systems Designer and Producer, founding team

Utah, USA | Aug 2026 - Present

- Translate design goals into implementation tasks and coordinate iteration. For Breach Brew, develop reusable Blueprint systems, parameter-driven environmental materials and Niagara feedback.

TimeStay | Part-time Systems and Balance Designer

Shanghai, China | 2026 - Present

- Contribute gameplay rules, numerical configuration and balance content; communicate implementation needs with the development team.

Selected Technical Projects

Punk Plush Panic | Unity capstone | 2025 - 2026

Improved boss attack timing and telegraphs; built configurable portals and path-driven traps; iterated from playtests and reduced scope for delivery. Released on Steam.

The E Book | Unity | Commercial project

Maintained gameplay-configuration spreadsheets and redesigned the data workflow; developed an AI-assisted Excel-to-Unity conversion tool.

Skills

Technical: Unreal Blueprint; basic Python and JSON; Excel configuration; Git and Perforce.

Simulation and content: project experience with Isaac Sim / Isaac Lab and RGB-D workflows; Unity, Godot, Niagara and material parameters; basic Maya / Blender modelling.

Languages: native Chinese; fluent English.

祁睿 Rui Qi

机器人与交互仿真方向

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机器人项目 | 游戏作品集

教育背景

犹他大学 University of Utah | Games 理学学士

2022.08 - 2026.04 | 学位授予日期 2026.04.30 | GPA 3.880/4.000

相关课程：Programming for All 1 / 2、统计推断导论、游戏引擎基础、传统游戏开发。多个学期获 Dean's List。

机器人项目

视觉抓取验证与失败恢复 | 个人仿真项目 | 2026.10

- 参与确定任务目标，反复运行演示，反馈抓空、空抓后继续运输及未重试等问题。质疑推开障碍是否必要，推动替代抓取方向的比较，以及后续真实堆叠任务。
- 项目结合颜色与深度定位、程序化抓取与侧抽、视觉载荷检查，以及抓错物体后的恢复。当前恢复逻辑由程序实现，并非新训练模型学到的能力。
- 当前流程与先搬红块的对照均完成 7/7 个开发布局；6 个堆叠场景中，2 个直接侧抽完成，4 个依靠恢复完成。平均耗时为 14.64 与 14.32 仿真秒，尚无独立最终测试结果。
- 另一项独立实验使用 PPO 训练小型神经网络，经过 96 回合学习选择三种抓取方式。在 12 个新开发验证布局中，模型与规则均完成 10/12，模型平均更慢（12.51 对 11.03 仿真秒）。

本人参与任务方向、演示测试与失败反馈、对照实验要求和结果检查；代码实现、调试及实验自动化使用了广泛的 AI 辅助。

相关工作经历

Qainbow LLC | 技术策划 / 系统策划 / Producer，创始团队

美国犹他州 | 2026.08 - 至今

- 将设计目标拆解为实现任务并协调迭代；参与 Breach Brew，开发可复用 Blueprint 系统、参数驱动的环境材质与 Niagara 反馈。

TimeStay | 兼职系统与数值策划

中国上海 | 2026 - 至今

- 参与玩法规则、数值配置与平衡内容设计，与开发团队沟通实现需求。

相关技术项目

Punk Plush Panic | Unity 毕业项目 | 2025 - 2026

调整 Boss 攻击时机并推动加入预警；制作可配置传送门与路径陷阱，根据试玩反馈迭代，控制项目范围以完成交付。项目已上线 Steam。

源书行者 The E Book | Unity 商业项目

维护玩法配置表，重新设计策划数据流程；使用 AI 辅助开发 Excel 到 Unity 数据格式的转换工具。

技能

编程与工具：Unreal Blueprint；基础 Python 与 JSON；Excel 配置；Git、Perforce。

仿真与内容：Isaac Sim / Isaac Lab 与 RGB-D 流程的项目实践；Unity、Godot、Niagara、材质参数；Maya / Blender 基础建模。

语言：中文母语，英语流利。